qaptiva-mbl

VERSION 0.1.0 | RESEARCH USER GUIDE | OCTOBER 2026

A minimal, reproducible workflow for many-body localization dynamics: submit Qaptiva AnalogQPU jobs, retrieve time-resolved autocorrelators, and benchmark against independent manual exact diagonalization.

1. Supported models

Imbrie (Pauli operators): $H = \Sigma\, \Delta_i\, Z_i\, Z_{i+1} + \Sigma\, (h_i\, Z_i + \Gamma\, X_i)$. Each bond Δ_i is sampled uniformly in [delta_min, delta_max], and each field h_i uniformly in $[-h0, h0]$.

Random-field XXZ (spin-1/2 convention): $H = \Sigma\, [Jxy\, (Sx_i\, Sx_{i+1} + Sy_i\, Sy_{i+1}) + Jz\, Sz_i\, Sz_{i+1}] + \Sigma\, h_i\, Sz_i$, where $S = \sigma/2$ and h_i is uniform in $[-h0, h0]$.

Users select **OBC/PBC**, the chain length L, Hamiltonian parameters, a random seed, and either a **Néel** or **random Z-basis product state**. Bit "0" has Z=+1; site 0 is the leftmost bit.

2. Observable

For the chosen product state, $s_j = \pm 1$. The measured autocorrelator is $\mathbf{C_j(t) = s_j\, \langle Z_j(t) \rangle}$; its site average is $\mathbf{C(t) = (1/L)\, \Sigma_j\, C_j(t)}$. Qaptiva returns raw Z_j magnetizations; the package applies s_j after retrieval.

This product-state autocorrelator is not automatically the infinite-temperature trace correlator.

3. Minimal capabilities

Qaptiva job ID persistence; lossless disorder arrays; independent NumPy/SciPy ED; site-by-site and site-average figures; error statistics and plots; full offline replotting.

No U(1) symmetry reduction, disorder sweeps, entanglement entropy, or noise models are included in v0.1.0.

Installation and settings

4. Install

From the repository root in a Python 3.10+ environment:

```
python -m pip install -e ".[notebook,test]"
python -m pytest -q
jupyter notebook notebooks/MBL_Simulation.ipynb
```

The `qat` and `qlmaas` modules are proprietary Qaptiva dependencies and are not bundled. You need a configured Qaptiva environment and account for remote operations. Offline ED and unit tests do not require Qaptiva.

5. Configure a sample

```
from qaptiva_mbl import RunConfig, prepare_run
cfg = RunConfig(model="imbrie", L=8, boundary="pbc",
                seed=12345, initial_state="neel", h0=3.5,
                gamma=1.0, delta_min=0.8, delta_max=1.2,
                tmax=100, error_control=True, nsteps=None,
                abs_tol=1e-8, rel_tol=1e-8, use_gpu=False)
run_dir = prepare_run(cfg)
```

For XXZ, set `model="xxz"` and choose `jxy` and `jz` (in the S=σ/2 convention). Each preparation creates a unique folder and saves exact input arrays.

6. Integrator choice

Adaptive: `error_control=True`; set positive `abs_tol` and `rel_tol`, and leave `nsteps=None`. The configured Qaptiva AnalogQPU chooses internal steps automatically using its adaptive Runge–Kutta Dormand–Prince DP5 method.

Fixed-step: `error_control=False` and provide `nsteps` as a positive integer; adaptive tolerances are not sent to the backend.

```
# Fixed-step alternative:
error_control=False, nsteps=1000
```

Internal solver steps and the measurement time grid are not interchangeable. The returned times are read directly from the remote results.

Jupyter notebook workflow

7. Prepare, submit, and retrieve

Open `notebooks/MBL_Simulation.ipynb`. Its remote cells are deliberately disabled by default; enable them only after configuration.

```
from qaptiva_mbl import submit_run, retrieve_run
job_id = submit_run(run_dir)  # requires Qaptiva access
print(job_id)
# Now, or from a later notebook session:
times, C_qpu = retrieve_run(run_dir,
                            host="YOUR-QLMAAS-HOST")
```

The remote job ID is loaded automatically from the saved `run.json`. QLMaaS manages login separately; never put credentials in a notebook or Git repository.

8. Manual exact diagonalization

```
from qaptiva_mbl import run_ed, compare_run
# ED uses saved disorder/state and exact QPU timestamps:
times, C_ed = run_ed(run_dir, max_L=12)
summary = compare_run(run_dir)
print(summary["global_max_absolute_error"])
```

Manual ED does not use QuSpin or Qaptiva operators. It supports dense diagonalization or sparse Krylov evolution. The default ED limit is L=12 to avoid excessive memory/time.

9. Plot locally

```
from qaptiva_mbl import plot_run
plot_run(run_dir, kind="sites", sites=[0,1,2,3],
         save="sites.pdf")
plot_run(run_dir, kind="average", save="average.pdf")
plot_run(run_dir, kind="error", save="errors.pdf")
```

Users may replot old jobs without resubmitting or reconnecting. Set `log_time=True` to use a logarithmic time axis.

Files and validation

10. Saved data

```
runs/imbrie_L8_<unique-id>/
  run.json              config, job ID, status, timestamps
  sample.npz            original delta bonds, fields, spins
  qpu_results.npz       times, magnetization, C_j(t), C(t)
  ed_results.npz        times, ED C_j(t), C(t)
  comparison.npz        signed/error arrays, RMSE, maxima
```

The package uses independent deterministic streams for the disorder and initial random state. Two prepared runs with the same seed and model but different state choices share the same disorder arrays.

11. How numerical differences are defined

Signed difference: $\Delta C_j(t) = C_j\text{^QPU}(t) - C_j\text{^ED}(t)$. **Absolute difference:** $|\Delta C_j(t)|$. Additional outputs include mean absolute error, RMS error and maximum absolute error across sites at each returned time, plus run-wide summary statistics.

ED and Qaptiva always use the same saved Hamiltonian arrays and state, and are compared only at matching times. The difference is numerical disagreement between solvers, not a finite-size or thermodynamic-limit error.

12. Useful checks

Import error? Use the site-supplied Qaptiva Python environment for remote calls. **Time-grid mismatch?** Rerun ED using QPU result times. **Unexpected XXZ scales?** XXZ uses spin-1/2 couplings while Imbrie uses Pauli couplings. **Submission uncertainty?** Inspect server job history before attempting a second submission.

Further information

Official Qaptiva analog overview | Getting started | Schedules

Status: automated offline tests and QLMaaS-shaped mock result parsing are checked; live remote submission/retrieval must still be verified on your Qaptiva installation.